

Instructions:

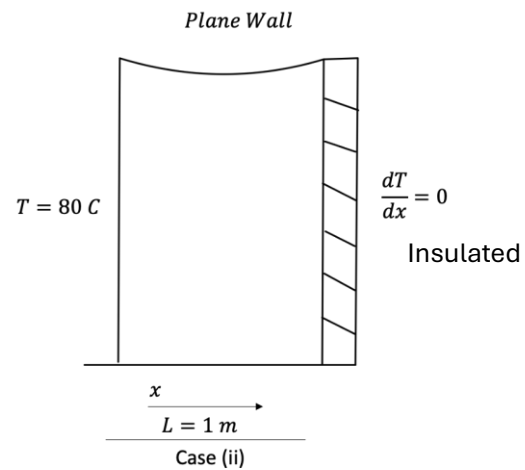
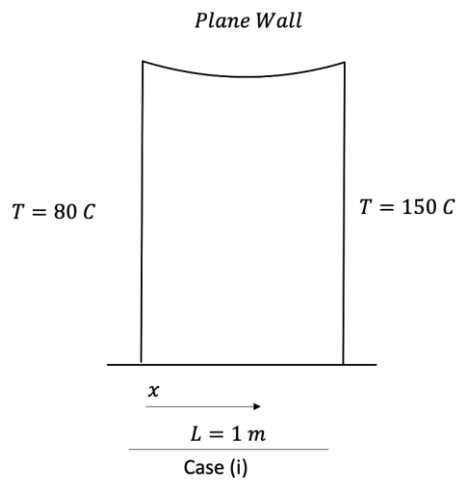
- Submit your assignment as a **single** PDF file through Canvas.
- Show all your steps and solution procedures.
- Make sure that your solution is well organised and that you are using appropriate headers for each question and sub-question.
- Scanned photos of your handwritten solution are acceptable as long as they are **legible**
- If a question requires plotting, please plot the expressions using any software/language (MATLAB, Excel, Mathematica, etc.) of your choice. However, **hand drawings** will not be accepted.
- You can use any software or calculator to perform any calculations. Using symbolic packages in MATLAB, Mathematica is acceptable as long as the solution steps are well laid out in your submission. In case you use those packages please attach the script in your submission file.
- While you can discuss the problems with classmates, all submitted work must be your own to conform to academic integrity guidelines.
- Late assignments will not be accepted.

Question 1 (20 points)

The heat transfer in a plane wall with no internal heat generation can be modelled using eq 1. Solve for the temperature $T(x)$ for the following two boundary conditions:

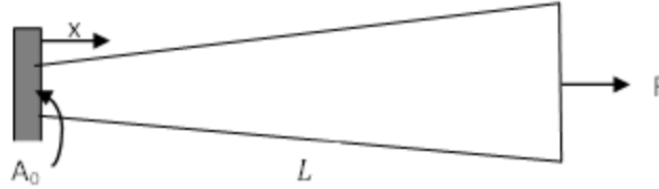
- Temperature specified at the two boundaries.
- Temperature specified at $x = 0$ and the temperature gradient $\frac{dT}{dx}$ is specified at $x = 1$

$$\frac{d^2T}{dx^2} = 0 \text{ (eq 1)}$$



Question 2 (30 points)

Consider a tapered bar of length L , Young's modulus E , and a variable cross-sectional area $A(x) = A_0 \left(1 + \frac{x}{L}\right)$. The bar is fixed from one end and a concentrated force F is acting on the other end. The axial displacement u can be modelled using eq 2. The body force is assumed to be zero in this problem.



$$-\frac{d}{dx} \left(EA \frac{du}{dx} \right) = b(x) \quad (\text{eq 2})$$

Show that the analytical solution is given by:

(30 Points)

$$u(x) = \frac{FL}{EA_0} \ln \left(1 + \frac{x}{L} \right)$$

Question 3 (50 Points)

The transverse displacement $y(x)$ of an Euler-Bernoulli beam can be expressed using eq 3.

$$EI \frac{d^4 y}{dx^4} = w(x) \text{ (eq 3)}$$

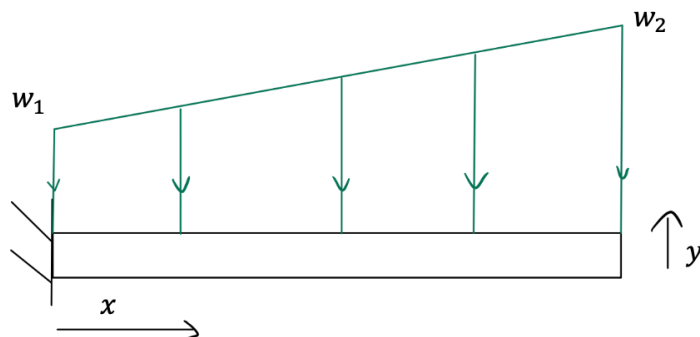
The relationship between the transverse displacement $y(x)$, the slope $\theta(x)$, shear force $V(x)$ and bending moments $M(x)$ are summarised in eq 4-6 below.

$$\theta(x) = \frac{dy}{dx} \text{ (eq 4)}$$

$$V(x) = EI \frac{d^3 y}{dx^3} \text{ (eq 5)}$$

$$M(x) = EI \frac{d^2 y}{dx^2} \text{ (eq 6)}$$

A cantilever beam is subjected to a distributed load as shown below.



- i. Find the expression for $w(x)$ given that
 - $w_1 = -50 \text{ N/m}$
 - $w_2 = -1000 \text{ N/m}$
- ii. Write the mathematical expressions of the boundary conditions.
- iii. Solve for $y(x)$ given that
 - $EI = 1 \times 10^6 \text{ Nm}^2$
 - $L = 1 \text{ m}$
- iv. Plot $y(x)$